Communication Engineering

Data and Signals:

1. Define a communication system and list its four requirements with brief explanations?

⇨ **Communication system:** Communication is the transfer of information from one place to another. A communication system is defined as a process of conveying information from a source to a destination. It should be efficient, reliable, and secured. An electronic communication system transfers information using an electrical field as a means of signal.

The four requirements are:

Rate of information transfer: The amount of info communicated per unit time, determines the physical form and technique used.

Simplicity of the system: The system must be convenient, effective, and easy to use.

Reliability: Users must be able to depend on it, it must work without errors or with acceptable error.

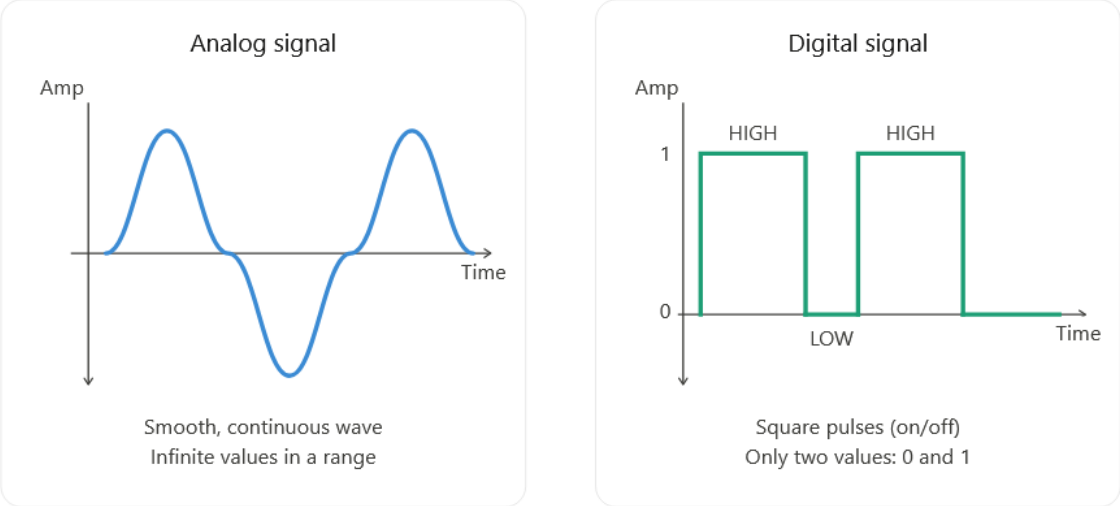
Purity of signal: The received signal must be the same as the transmitted signal.

2. Compare Analog and Digital communication systems, discussing their respective advantages and disadvantages with a diagram illustrating their signal shapes?

⇨ Analog signals are continuous they can take an infinite number of values within a range. In the time domain, they appear as smooth sinusoidal waveforms. An analog signal carries all information from the original source without any sampling or approximation.

Digital signals are discrete they take only specific values (0 and 1). In the time domain, they appear as rectangular square pulses transitioning abruptly between a high and a low state. A digital system samples the original signal at intervals and converts each sample into a binary number.

Feature	Digital Communication System	Analog Communication System
Data merging	Different data types (voice, video, text) can be merged and multiplexed on a single channel.	Cannot merge different types of data on the same channel.
Error correction	Supports error detection and correction (e.g., parity, CRC, Hamming codes).	No error correction capability.
Bandwidth required	Requires larger bandwidth to carry the same information.	Requires smaller bandwidth.
Synchronization	Synchronization is relatively more difficult; requires clock recovery circuits.	Synchronization is relatively easier.
Application examples	Mobile network (4G/5G), Internet, digital TV, VoIP.	AM/FM radio, traditional telephone (PSTN), analogue TV.

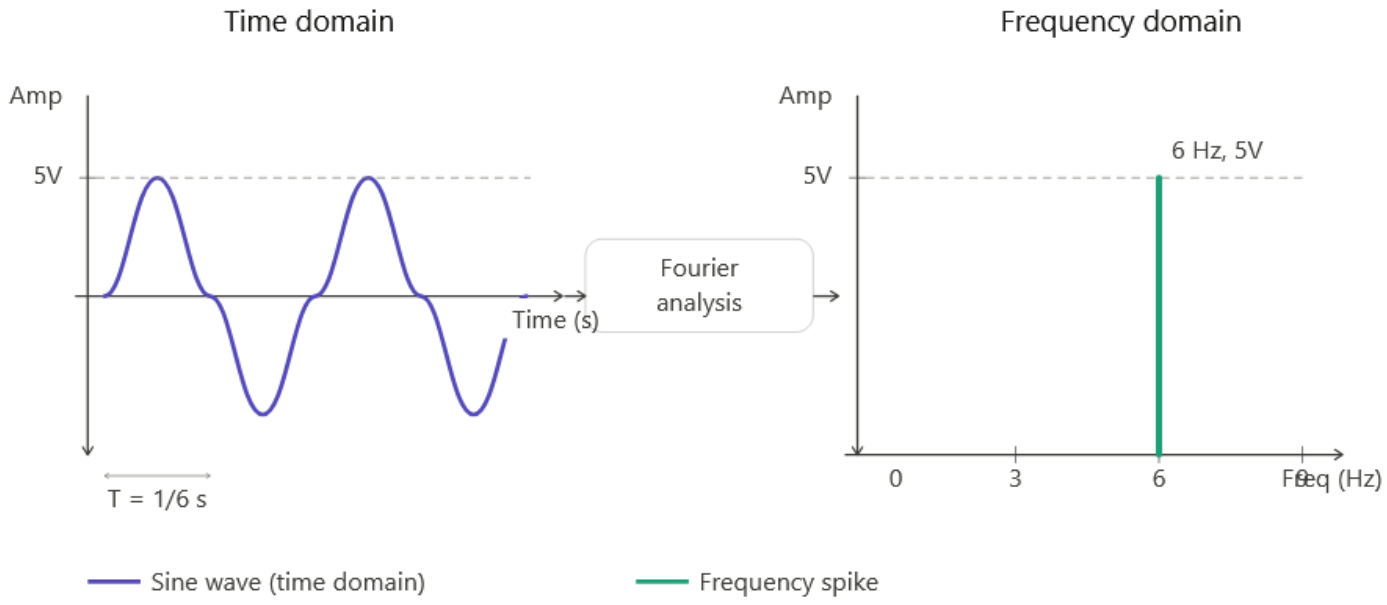


3. Explain the difference between the time domain and frequency domain of a signal. How can a time domain sine wave be converted to the frequency domain? Use a diagram.

⇨ Time domain shows how the amplitude of a signal changes over time. The X-axis represents time and the Y-axis represents amplitude. It gives us a direct visual of the signal's shape and behavior. For example: A sine wave oscillating up and down.

Frequency domain shows which frequencies are present in a signal and what amplitude each frequency carries. The X-axis represents frequency (Hz) and the Y-axis represents amplitude. Instead of showing how the signal varies over time, it shows the signal's frequency composition.

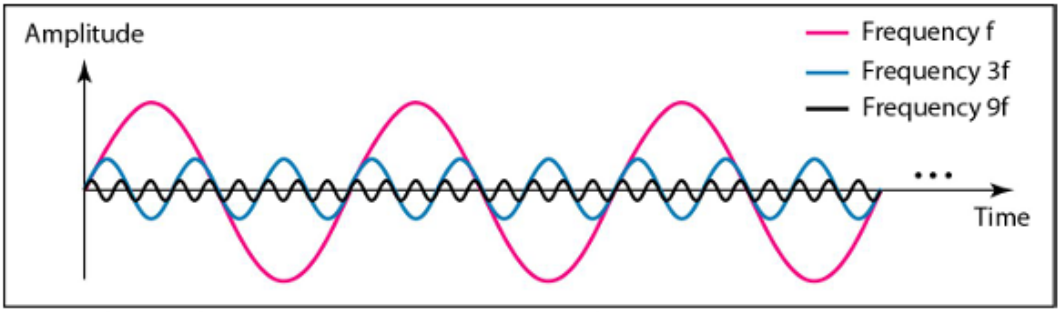
Time Domain to Frequency Domain Conversion:



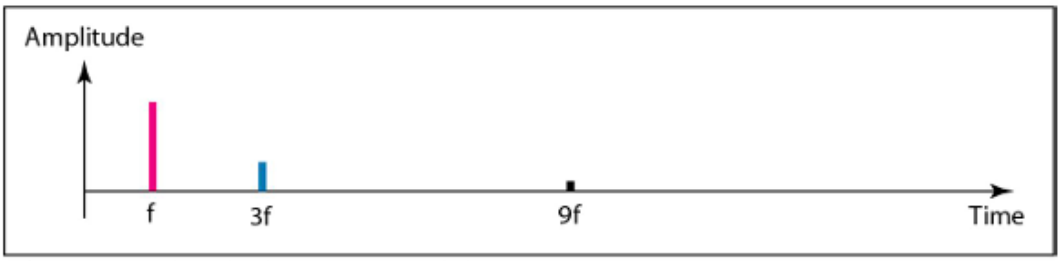
A complete sine wave in time domain = one single spike in frequency domain
(Forouzan, Figure 3.7)

According to Figure 3.7 a sine wave with a peak value of 5V and a frequency of 6 Hz appears as a continuous oscillating wave in the time domain. When converted to the frequency domain using Fourier analysis, that same signal appears as a single vertical spike at 6 Hz with amplitude 5V.

Figure 3.10 ***Decomposition of a composite periodic signal in the time and frequency domains***



a. Time-domain decomposition of a composite signal



b. Frequency-domain decomposition of the composite signal

4. Define bandwidth. A composite periodic signal is decomposed into components with frequencies of Pink → f, Blue → 3f, Black → 9f. If f = 1000 Hz, calculate the bandwidth and draw the frequency domain diagram. Also explain the difference between a simple signal and a composite signal?

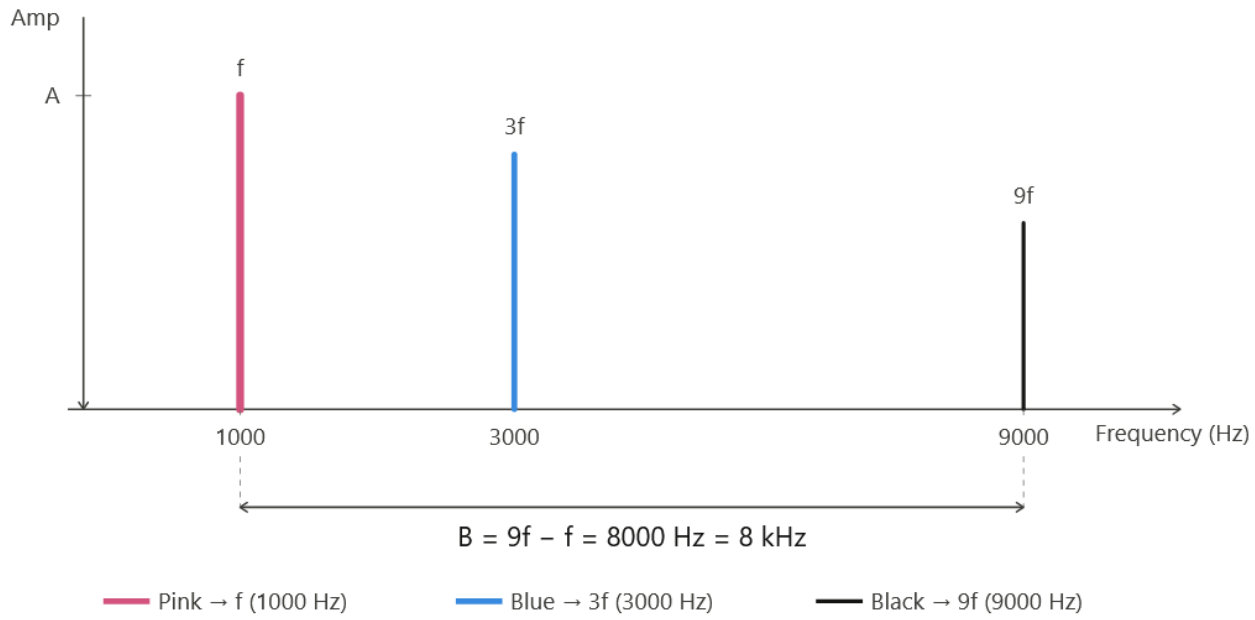
⇨ The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

Formula: $B = f_{high} - f_{low}$

Given, Pink → frequency f, Blue → frequency 3f, Black → frequency 9f, f = 1000 Hz

f = 1000 Hz 3f = 3000 Hz 9f = 9000 Hz

Bandwidth calculation: $B = f_{high} - f_{low} = 9000 - 1000 = 8000 \text{ Hz} = 8 \text{ kHz}$

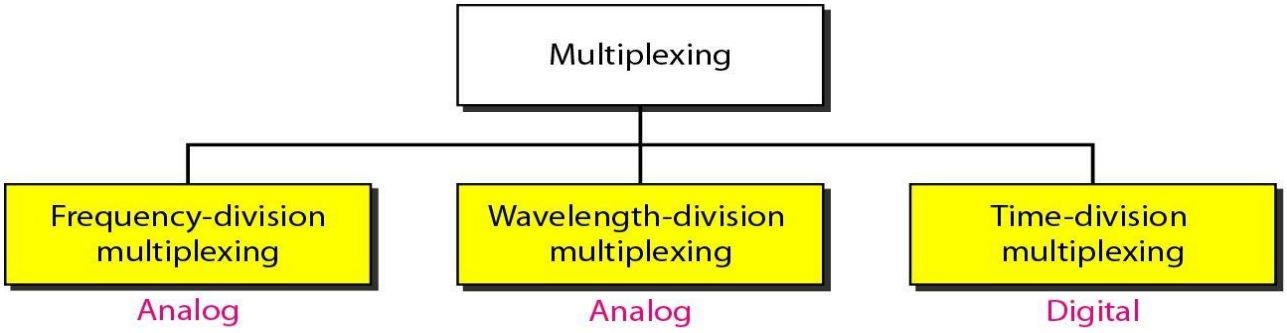


The three spikes appear at f, 3f, and 9f each representing one sine wave component of the composite periodic signal. The spike heights reflect their respective amplitudes (Pink is the largest, Black the smallest), and the horizontal span between the lowest (1000 Hz) and highest (9000 Hz) frequency gives the bandwidth = 8000 Hz.

Multiplexing

1. What is multiplexing? Explain its categories. (2 marks)

⇨ Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link. This works because whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of those devices that extra capacity does not need to sit idle. The link can be shared among many devices instead. As data and telecommunication use keeps growing traffic also increases so sharing one link among many signals saves cost and uses bandwidth wisely.

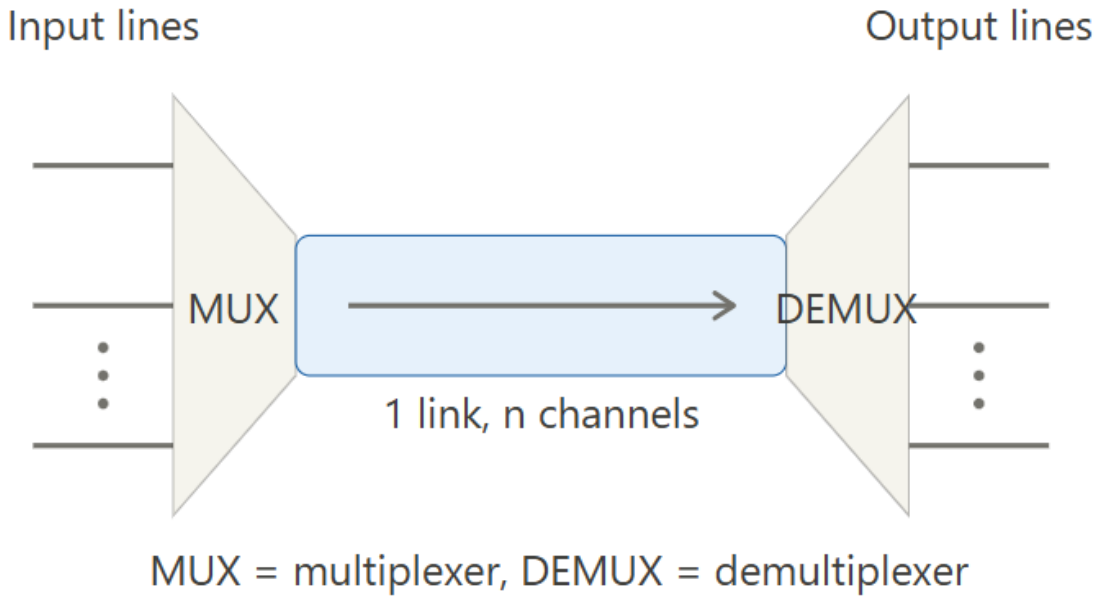


2. Describe the frequency position of multiplexing and Demultiplexing? With example?

⇨ Frequency position means the separate frequency band given to each signal so multiple signals can travel on one link without mixing.

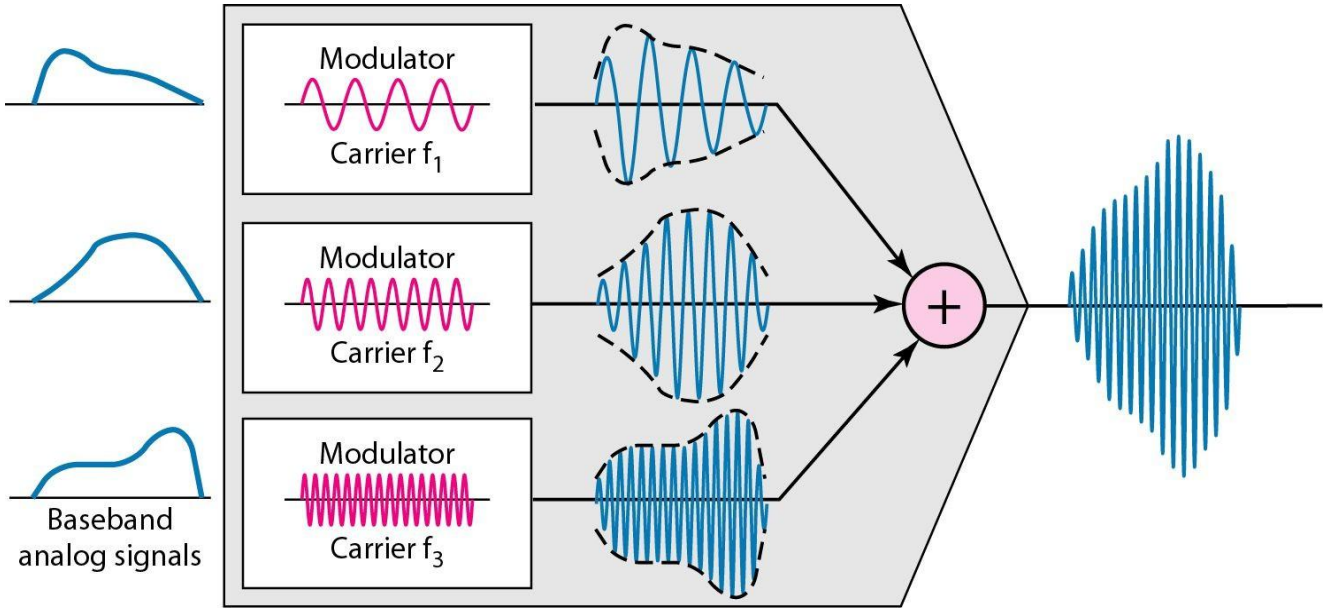
Multiplexing (MUX): Each input baseband signal is passed through its own modulator with a different carrier frequency (f1, f2, f3...). This shifts each signal to a separate, non-overlapping frequency band. All shifted signals are then added together and sent on one link.

Demultiplexing (DEMUX): At the receiver, band pass filters tuned to each frequency band separate the combined signal back into individual carrier signals. A demodulator then shifts each one back to its original baseband range.



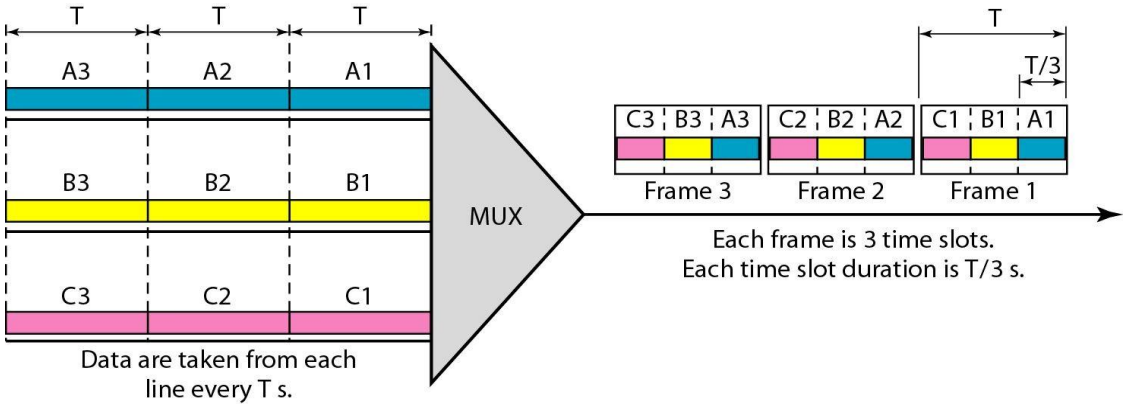
3. Explain the FDM process with diagram. (3 marks)

⇨ FDM is an analog technique. Each baseband signal is fed into its own modulator and each modulator uses a different carrier frequency (f1, f2, f3) so every signal gets its own frequency band without overlapping. A combiner adds all the modulated signals into one composite signal which is sent over the link. At the receiver filters separate the composite signal back into its frequency bands and demodulators recover each baseband signal using the matching carrier.



4. What is TDM? Explain Synchronous TDM with example. (2 marks)

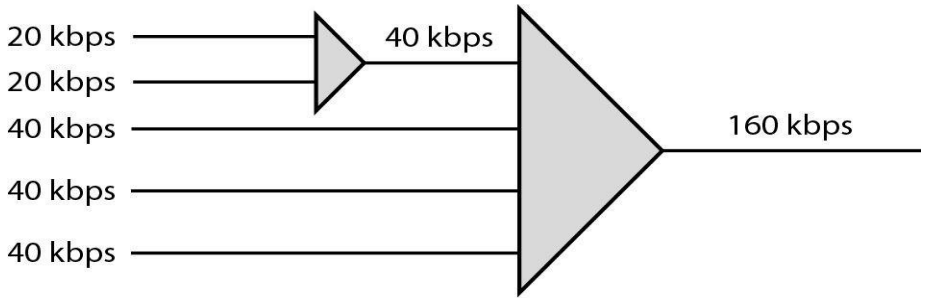
⇨ TDM is a digital technique that combines many low-rate channels into one high-rate channel by sharing time, not frequency. In Synchronous TDM, every input gets exactly one slot in every frame, turn by turn, even if that input has nothing to send. If there are n input lines, the link's data rate becomes n times faster, and each slot lasts only T/3 of the original time.



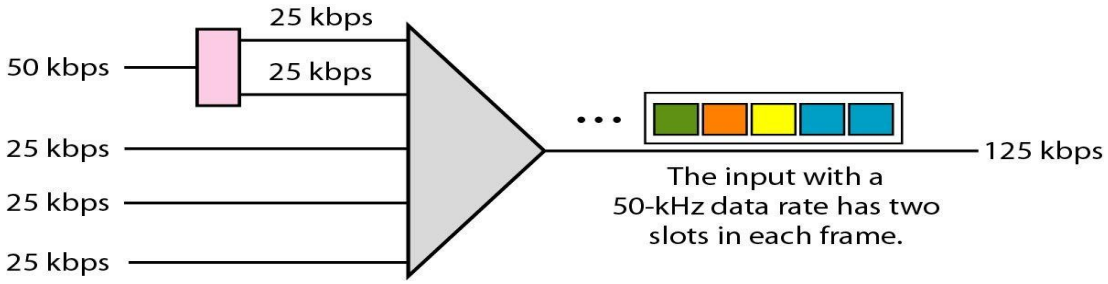
5. What is Data Rate Matching? Explain Multilevel, Multi slot, and Pulse Stuffing. (5 to 10 marks).

⇨ Data Rate Matching means fixing the mismatch when input links going into a multiplexer have different data rates, using one of three strategies: multilevel, multi slot, or pulse stuffing.

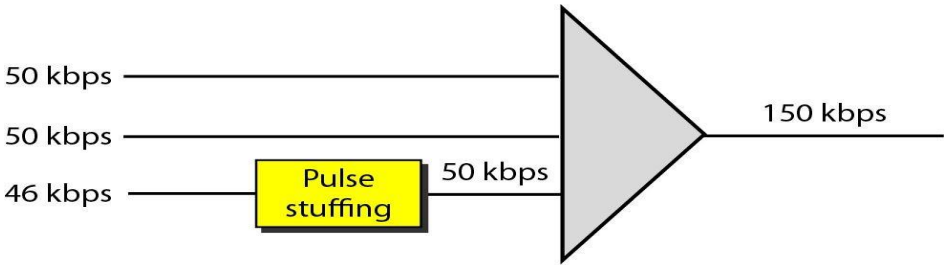
Multilevel: Used when the data rate of the input links are multiples of each other.



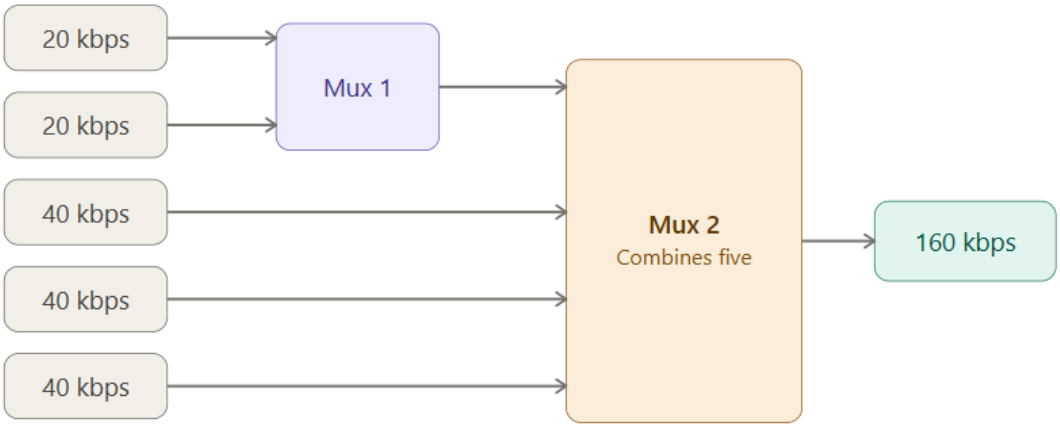
Multi slot: Used when there is a GCD between the data rates. The higher bit rate channels are allocated more slots per frame, and the output frame rate is a multiple of each input link.



Pulse Stuffing: Used when there is no GCD between the links. The slowest speed link will be brought up to the speed of the other links by bit insertion, this is called pulse stuffing.



6. **The circuit below shows five input links with different data rates being combined into one link. Identify the data rate matching technique used in this circuit and explain how it works?**



⇨ The data rate matching technique used in this circuit is Multiple-Slot Multiplexing, which is a strategy within Synchronous Time-Division Multiplexing (TDM) applied when input links have different data rates that share a Greatest Common Divisor (GCD).

Multiple-slot multiplexing circuit

